

M52 — Repeating alarm audio + 45 s silence default + brand rename

Phase: Phase 7 follow-on (post-M51, operator feedback) **Status:** **DONE** **Blocked by:** M49 (alarm sounds), M50 (silence), M51 (PIA cascade), M41 (print sheet) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Three asks from the operator, bundled:

"Repeat alarm sounds until cleared, silence of an alarm last 45 seconds then it goes active if the condition is not resolved or cleared. Branding on screens and printed material- 'Training Bridge VRAI- MedSim'"

Delivered:

1. **Repeating alarm audio** on the Nursing Station and on the PIA cascade. M49 played each alarm's WAV exactly ONCE on first sight (good enough for a demo but unsafe for a real drill — the operator could miss the only chime). Now the audio dispatcher keeps a per-alarm last-played timestamp and replays on a cadence matched to the priority field already on every alarm: - **high** (critical + danger): every 8 s - **medium** (warning): every 20 s - **low** (info): every 45 s PIA's code-blue cascade poller uses a fixed 8 s repeat. Silenced alarms skip the play call but the timestamp tracking continues. 2. **45 s silence default**. M50 shipped silence with a 120 s default; operator wants 45 s so the alarm gets a chance to re-fire if the underlying breach hasn't resolved. The route's `?seconds=N` override still works for the rare case an operator wants longer. 3. **Brand rename**. Every user-visible screen title + the printable QR sheet now reads **"Training Bridge VRAI- MedSim"** (exact spacing per operator). Code-level identifiers (`window.MEDSIM2_OPS`, `~/medsim/` filesystem path, `medsim_ehr_window` browser target) are left alone — those are code, not user-visible brand.

2. Files touched

Server

- `portal/alarms.py` — `silence_alarm` helper default `duration_s 120` → 45.
- `portal/server.py` — `POST /api/alarm/{id}/silence` route `seconds: int = 120` → 45. Comment updated.

Client JS

- `portal/static/nurse_station.js`
- Replaced `_seenAlarmIds`: Set with `_audioLastAt`: Map<alarm_id, ms>.
- Added `AUDIO_REPEAT_MS = {high: 8000, medium: 20000, low: 45000}`.
- Re-write `_playNewAlarmSounds(alarms)` to consult `audio_priority`, pick the cadence, replay when `(now - last) ≥ cadence`, skip silenced alarms, and drop ids that leave the active list.
- Silence button title now reads "Mute audio for 45 s without clearing the breach (audio resumes after if still active)".
- `portal/static/pia_app.js`
- Added `CASCADE_AUDIO_REPEAT_MS = 8000 + _cascadeAudioLastAt`: Map.

- Cascade poller now plays 03_code_blue.wav every 8 s per active code-blue alarm (gated per alarm_id), while keeping the _cascadeKey flash dedupe so the CSS animation doesn't restart on every poll.

Templates (brand rename)

- portal/templates/base.html — <title> and topbar brand text.
- portal/templates/home.html — <h1>.
- portal/templates/login.html — <title> + <h1>.
- portal/templates/join.html — <title> + <h1>.
- portal/templates/ehr_join.html — <title>.
- portal/templates/device_app.html — <title> + apple-mobile-web-app-title meta.
- portal/templates/device_join.html — <title> + meta.
- portal/templates/device_pia.html — <title> suffixed with brand.
- portal/templates/nurse_station.html — <title> suffixed with brand.
- portal/templates/qr_print.html — <title> + .brand-title h1 (two occurrences).

Tests

- tests/v7/test_repeat_audio_silence_brand.py (new) — 18 acceptance tests covering all three asks.
- tests/v7/test_clinical_alarm_sounds.py — test_nurse_station_js_deduces_by_alarm_id was asserting the old once-per-occurrence behaviour. Renamed to test_nurse_station_js_repeats_audio_by_cadence and re-pointed at the new _audioLastAt / AUDIO_REPEAT_MS symbols.
- tests/v7/test_qr_print_sheet.py — brand assertion swapped from "Training Bridge MedSim-VRAI" to "Training Bridge VRAI- MedSim".

3. Acceptance — what M52 must satisfy

Source: tests/v7/test_repeat_audio_silence_brand.py (18 tests).

3.1 Silence default

- POST /api/alarm/{id}/silence with no ?seconds query lands at 45 s. Response body says duration_s == 45 and silenced_until ≈ now + 45.
- POST /api/alarm/{id}/silence?seconds=120 still works — operator override preserved.
- inspect.signature(alarms.silence_alarm).parameters["duration_s"].default == 45.

3.2 Repeating audio on the Nursing Station

JS source must carry:

- _audioLastAt (Map identifier).
- AUDIO_REPEAT_MS with high, medium, low keys.
- Reads audio_priority from each alarm dict.
- Silenced alarms still skipped (a.silenced).

- Map cleared when alarms list goes empty.
- Old `_seenAlarmIds` symbol completely gone (code AND comments).

3.3 Repeating audio on the PIA cascade

JS source must carry:

- `_cascadeAudioLastAt` Map.
- `CASCADE_AUDIO_REPEAT_MS` constant set to 8000.
- Silenced alarms skipped.
- Map cleared when no code-blue alarms are active.
- `_cascadeKey` retained — flash CSS still gated by NEW alarm sets.
- `playSound('code_blue')` still wired.

3.4 Brand on every user-visible screen

- `base.html` contains "Training Bridge VRAI- MedSim" and no "MEDSIM 2".
- Same check for `home.html`, `login.html`, `join.html`, `ehr_join.html`, `device_app.html`, `device_join.html`, `device_pia.html`, `nurse_station.html`.
- `qr_print.html` contains the brand string ≥ 2 times (title + visible header) and zero occurrences of the old "Training Bridge MedSim-VRAI".
- End-to-end HTTP fetch of `/portal/home` and `/portal/control/qr_print` includes the new brand and excludes the old one.

All 18 tests pass; full v7 suite **405 passed, 1 skipped, 0 regressions** (was 387 pre-M52).

4. Cadence values — why these numbers

The cadence table mirrors real bedside-monitor behaviour:

Priority	Cadence	Source
high	8 s	Critical/danger alarms — short repeat
medium	20 s	Warning alarms — moderate
low	45 s	Info alarms (call bell) — gentle

The PIA cascade fires at the high rate (8 s) because every code blue is by definition top-priority. Other PIA presses (`call_bell`, `bed_alarm`) don't go through the cascade poller — they show as device alarms on the nurse station and get their own per-priority cadence there.

Why not just play continuously? Bedside students need to be able to talk over the alarm, communicate with the team, and hear the nurse station's intercom. A short repeating tone (rather than a continuous one) leaves space for clinical communication while keeping the alarm psychologically present.

Interaction with silence: a silenced alarm doesn't tick its last-played timestamp forward — but the silence itself expires after 45 s by default (down from M50's 120 s). So a worst-case sequence is:

1. SpO2 dips → audio plays at $t=0$.
2. Operator silences → audio off, alarm board shows ■.
3. 45 s later, silence expires; if breach still active, the audio dispatcher sees it un-silenced again on the next poll and re-fires at the high cadence.

The operator's stated requirement — `""silence of an alarm last 45 seconds then it goes active if the condition is not resolved or cleared""` — is satisfied because the silence-record TTL and the audio-loop repeat are now both wired in.

5. What was deliberately NOT renamed

- `window.MEDSIM2_OPS`, `window.MEDSIM2_STATION`, `window.MEDSIM2`, `window.MEDSIM_SESSION` (JS globals — code-level identifiers referenced from .js source files; renaming would silently break the bootstrap).
- `~/medsim/vault.enc` (filesystem path — renaming would lose every existing user's vault).
- `target="medsim_ehr_window"` (browser-window target name; matches across multiple links so we don't open three EHR tabs).
- `var KEY = 'medsim_a2hs_dismissed_at'` (localStorage key — code).
- `"Voice4MedSim_v6"` historical attribution in personas/curriculum docs (correct attribution of the source).
- `BUILD_STATE.md` and `MODULE_GUIDE_TEMPLATE.md` body text (internal engineering docs, not user-facing).

If the operator wants any of those renamed too they're a separate ticket — each one touches code paths that need migration.

6. Demo script for the operator

1. Hit `/portal/home` — top-left brand should read **Training Bridge VRAI- MedSim**, browser tab title same. Old "MEDSIM 2" gone. 2. From an active room, click "Print all encounters" — printed sheet header says **Training Bridge VRAI- MedSim** in the top-of-page banner. 3. Force an SpO2 breach (Per-Patient Console → set SpO2 = 80). Within 3 s the Nursing Station should: - Play the high-priority chime. - Re-play it every ~8 s until cleared. 4. Click ■ Silence on that alarm. The chime stops. Wait 45 s. The audio comes back on its own. 5. Click Clear. The chime stops and the alarm disappears. 6. Register a PIA on Bed 1, another on Bed 2. From Bed 1's PIA, hit Code Blue. Both PIAs flash red and play the code-blue tone every ~8 s until the alarm is cleared from the nurse station.

Closes: the operator's three asks from the M52 message in full. No follow-on tickets identified.